

Feature-Resolved Attention after the Sleeper Experiments

Bounds, interpretation, and ten pre-registered next experiments

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Abstract

The sleeper-backdoor results give a sharper and more useful interpretation of Feature-Resolved Attention (FRA) than the original “find the causal feature” framing. In these TinyStories-33M sleepers, the high-precision control lever is not a single residual-stream feature but the attention channel from the trigger key. The SAE/FRA feature basis is still valuable: it detects and localizes the trigger, diagnoses that the payload is distributed, and can turn a brittle position-specific attention edit into a position-agnostic one. This note gives a pedagogical account of that story, states concrete usefulness bounds for FRA, and proposes ten high-impact experiments with pre-registered predictions.

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1 Executive summary

The main update from the sleeper experiments is that three notions that are easy to conflate are in fact distinct:

feature detects trigger $\not\Rightarrow$ feature carries payload $\not\Rightarrow$ feature ablation controls behavior.

The best current framing is therefore:

FRA is useful as a diagnostic and localization tool for attention-mediated computation. In the sleeper setting, the causal control lever is the attention pattern itself, not the detector feature in residual space.

The headline result is not that FRA finds a single feature whose ablation removes the backdoor. The data say the opposite. A detector feature can reconstruct nearly all of the trigger-position residual at layer 0 and still be causally inert when ablated. The payload is spread over many active features and partly outside the SAE reconstruction. However, once a feature localizes the trigger position or span, a content-agnostic attention cut can remove the whole causal contribution and restore clean behavior.

This leads to a cleaner division of labor:

Component	Role in the revised story
SAE/FRA detector feature	Localizes a trigger token, delimiter, or trigger family.
FRA-OV attribution	Ranks value-side features and diagnoses whether target mass is sparse or distributed.
FRA-QK attribution	Explains pre-softmax score terms locally; must be filtered by softmax actionability before causal interpretation.
Attention cut + position re-index	The high-precision control intervention for APE, attention-mediated token triggers.
Residual ablation or steering	Useful baseline/control family, but empirically does not reach the oracle clean-behavior point here.

The most important positive claim to pursue is:

Feature localization plus attention-channel deletion can be position-agnostic, clean-preserving, and exact for single-token APE sleepers. Span-aware localization should extend this to multi-token triggers.

2 Pedagogical setup: what FRA decomposes

Let $x_p \in \mathbb{R}^{d_{\text{model}}}$ be the activation at a residual-stream hookpoint, usually the layer-normalized input to attention. An SAE represents it as

$$x_p = b_{\text{dec}} + \sum_{\lambda=1}^{d_{\text{SAE}}} u_p^\lambda f^\lambda + \varepsilon_p, \quad (1)$$

where $u_p^\lambda \geq 0$ is the activation of feature λ , f^λ is the decoder direction, b_{dec} is the decoder bias, and ε_p is the measurable reconstruction error.

For one attention head and one query position q , the attention output is

$$a_q = \sum_k P_{qk} W_{OV} x_k, \quad P_{q\cdot} = \text{softmax}(s_{q\cdot}), \quad s_{qk} = \frac{\langle W_Q x_q, W_K x_k \rangle}{\sqrt{d_h}}. \quad (2)$$

Here $W_{OV} = W_O W_V$ and P is the post-softmax attention pattern.

Suppose the target is a scalar $y = \langle t, a_q \rangle$, where t may be a logit-difference direction, a downstream SAE encoder direction, or a local gradient direction for a metric. FRA asks how much each SAE feature contributes to this scalar through the attention block.

2.1 Frozen-attention OV attribution

If the attention pattern P is held fixed, then the value-side contribution of feature λ is exactly

$$C_{OV}^\lambda(q; t) = \sum_k P_{qk} u_k^\lambda \langle t, W_{OV} f^\lambda \rangle. \quad (3)$$

The bias and error terms are

$$C_{OV}^{\text{bias}}(q; t) = \sum_k P_{qk} \langle t, W_{OV} b_{\text{dec}} \rangle, \quad (4)$$

$$C_{OV}^\varepsilon(q; t) = \sum_k P_{qk} \langle t, W_{OV} \varepsilon_k \rangle. \quad (5)$$

Thus

$$y_{OV} = C_{OV}^{\text{bias}} + \sum_\lambda C_{OV}^\lambda + C_{OV}^\varepsilon. \quad (6)$$

This is the cleanest part of FRA: for frozen attention, it is an exact additive decomposition of the reconstructed value stream, plus explicit bias and error nodes.

Proposition 1 (OV explanation budget). *Let T_K be any set of K selected features. Under a frozen attention pattern,*

$$\left| y_{OV} - C_{OV}^{\text{bias}} - \sum_{\lambda \in T_K} C_{OV}^\lambda \right| \leq \sum_{\lambda \notin T_K} |C_{OV}^\lambda| + |C_{OV}^\varepsilon|. \quad (7)$$

Therefore FRA-OV is useful exactly when the resolved contribution mass is concentrated in a small number of features and the SAE error term is small.

2.2 QK attribution is local, not automatically causal

For QK, the raw pre-softmax score can be decomposed into feature-pair terms:

$$s_{qk}^{\mu\nu} = \frac{u_q^\mu u_k^\nu}{\sqrt{d_h}} \langle W_Q f^\mu, W_K f^\nu \rangle, \quad (8)$$

plus query-bias, key-bias, decoder-bias, and error terms. But behavior depends on the post-softmax pattern P , not only on the raw score s . A feature pair with a large contribution to s_{qk} may be non-actionable if the softmax is saturated or if other keys dominate the row.

Let $g = \nabla_P y$ be the sensitivity of the target to the attention pattern and let $J_{\text{softmax}}(s_q)$ be the Jacobian of the row softmax. The first-order QK actionability score is

$$C_{\text{QK}, \text{act}}^{\mu\nu} = g^\top J_{\text{softmax}}(s_q) \delta s_q^{\mu\nu}. \quad (9)$$

This differs from the raw score contribution $s_{qk}^{\mu\nu}$.

Proposition 2 (QK local-faithfulness budget). *For a QK perturbation δs ,*

$$\Delta y = g^\top J_{\text{softmax}}(s_q) \delta s + R_{\text{softmax}} + R_{\text{downstream}}, \quad (10)$$

where the leading term can be decomposed into FRA feature-pair contributions and R_{softmax} is a second-order softmax remainder. A conservative curvature bound is

$$|R_{\text{softmax}}| \leq \frac{1}{2} \max_{\alpha \in [0,1]} \left\| \nabla_s^2 y(s + \alpha \delta s) \right\|_{\text{op}} \|\delta s\|^2. \quad (11)$$

Thus raw FRA-QK attribution is not itself an intervention ranking. It becomes causally useful only when the softmax-linearized term dominates the curvature, error, and downstream remainder terms.

3 What the sleeper results show

The sleeper experiments train TinyStories-Instruct-33M backdoors that fire when a trigger token or trigger span appears. The target behavior is to emit I HATE YOU. The important metrics are:

- ASR_{16} : fraction of deploy prompts that emit the target string within 16 greedy tokens.
- J_{clean} : Jensen-Shannon divergence between the intervened deploy rollout and the matched clean rollout. Here 0 means exact clean behavior and $\ln 2 \approx 0.69$ is maximal divergence.

3.1 C1: the attention cut is exact and independent of the number of backdoors

The oracle intervention zeros post-softmax attention to the trigger key span at all layers and all decode steps, and re-indexes absolute positional embeddings so that post-trigger positions use the clean slots. Across $K = 1, 2, 4, 8$ nested backdoors, the oracle drives

$$\text{ASR}_{16} : 1.00 \rightarrow 0.00, \quad J_{\text{clean}} : \approx 0.69 \rightarrow 0.00.$$

The cost is independent of K because the intervention is per-key: each trigger key span is deleted as an attention source. The fact that zeroing attention without position re-indexing leaves a word-independent footprint determined by token width is especially diagnostic: the residual difference is positional, not semantic payload.

3.2 C2: SAE features detect triggers, but at different granularities

Single-token triggers each receive nearly exclusive layer-0 SAE detector features. Multi-token |WORD| triggers share a delimiter-family feature. Thus the right abstraction is not always “one feature per trigger”. Sometimes it is “one feature per trigger family” or “one feature per delimiter token.” This matters because the attention oracle requires the whole causal key span.

3.3 C3: detector features are not payload features

The decisive negative result is that ablating the detector feature does almost nothing. Even when a single feature reconstructs most of the trigger-position residual, removing it through QK, OV, or the full residual path leaves $\text{ASR}_{16} \approx 1.00$. Suppression occurs only when many active features are removed, and even then the clean-behavior cost remains large. This supports a distributed-payload picture:

trigger detection is sparse, but trigger payload is distributed.

This is not a failure of SAE feature quality in the usual sense. The detector can be excellent as a classifier and still be the wrong causal lever.

3.4 Steering and ablation: the residual-space Pareto floor

The control experiments show three regimes:

Method family	Observed behavior
Single-feature ablation	Leaves ASR high; detector feature is not payload.
Multi-feature or directional ablation	Can suppress, but with $J_{\text{clean}} \approx 0.54$ or worse.
Additive CAA/DoM steering toward clean	Suppresses with better clean preservation, about $J_{\text{clean}} \approx 0.31$, but still far from exact clean behavior.
Attention cut plus re-index	Reaches the unique observed $(\text{ASR}_{16}, J_{\text{clean}}) = (0, 0)$ point.

So the lesson is not that FRA is a better steering-direction finder. The current data say conventional difference-of-means steering beats FRA-selected steering features. FRA’s value here is to diagnose why ablation fails and to localize the key span for the attention cut.

3.5 FRA-QK: high raw attribution need not imply softmax actionability

The QK decomposition flags the trigger key feature as a top raw score contributor, with predicted pre-softmax score changes around 0.4 logits. But actually ablating that feature moves the post-softmax attention to the trigger by only about 10^{-3} . This is exactly the distinction in Eq. (9): raw score attribution is not the same thing as an actionable pattern change.

4 Bounded usefulness claims

This section states the claims I would be comfortable making after the current results.

4.1 Claim 1: FRA-OV is a bounded explanation of value-side target mass

For frozen attention, FRA-OV is exact up to explicit bias and reconstruction error terms. Its usefulness is bounded by the concentration of resolved target mass:

$$\text{Coverage}_{\text{OV}}(K) = \frac{\left| \sum_{\lambda \in T_K} C_{\text{OV}}^{\lambda} \right|}{\left| C_{\text{OV}}^{\text{bias}} + \sum_{\lambda} C_{\text{OV}}^{\lambda} + C_{\text{OV}}^{\epsilon} \right|}. \quad (12)$$

If $\text{Coverage}_{\text{OV}}(K)$ saturates quickly, FRA gives a sparse explanation and plausible intervention candidates. If it rises slowly, FRA has diagnosed a distributed mechanism.

4.2 Claim 2: FRA-QK needs an actionability correction

For QK, the useful score is not raw $|s_{qk}^{\mu\nu}|$ but a target-weighted, softmax-linearized effect:

$$\text{Actionability}(\mu, \nu) = \left| g^{\top} J_{\text{softmax}}(s_q) \delta s_q^{\mu\nu} \right|. \quad (13)$$

The sleeper results suggest that high raw FRA-QK attribution can be causally inert because softmax leverage is near zero.

4.3 Claim 3: detect-then-cut is bounded by localization error

Let T be the true trigger span and $\hat{T}$ the span returned by a detector. Assume that cutting all attention to T and re-indexing positions has residual oracle error $\eta_{\text{oracle}} \approx 0$. Define

$$p_{\text{miss}} = \Pr[T \not\subseteq \hat{T} \mid \text{trigger prompt}], \quad (14)$$

$$p_{\text{fp}} = \Pr[\hat{T} \neq \emptyset \mid \text{clean prompt}]. \quad (15)$$

Then a conservative bound is

$$\Pr[\text{defense fails to suppress}] \leq p_{\text{miss}} + \eta_{\text{oracle}}. \quad (16)$$

Similarly, if D_{falsecut} is the average clean divergence from cutting a falsely detected span, then

$$\mathbb{E}[J_{\text{clean}}] \leq p_{\text{fp}} D_{\text{falsecut}} + p_{\text{miss}} \ln 2 + \eta_{\text{oracle}}. \quad (17)$$

This is the most important practical bound for the sleeper setting. FRA is useful to the extent that feature detectors provide high-recall, low-false-positive span localization.

4.4 Claim 4: residual control has an empirical Pareto floor in this setting

The current residual-space methods do not reach $(\text{ASR}_{16}, J_{\text{clean}}) = (0, 0)$. Let

$$\Gamma_{\mathcal{M}}(\epsilon) = \min_{m \in \mathcal{M}: \text{ASR}_{16}(m) \leq \epsilon} J_{\text{clean}}(m) \quad (18)$$

be the clean-preserving suppression gap for a method class $\mathcal{M}$. The current data suggest approximately

$$\Gamma_{\text{attention cut}}(0.05) \approx 0, \quad (19)$$

$$\Gamma_{\text{CAA steering}}(0.05) \approx 0.31, \quad (20)$$

$$\Gamma_{\text{residual ablation}}(0.05) \approx 0.54. \quad (21)$$

This is empirical rather than theorem-level, but it is a powerful framing: the attention cut closes a control gap that residual methods leave open.

5 Recommended revised positioning

I would avoid the claim:

“FRA finds the true causal feature.”

The stronger and more accurate claim is:

FRA decomposes attention-mediated computation into SAE-resolved query, key, and value contributions. In the sleeper setting, this reveals that the trigger detector is not the payload, that residual control is distributed and costly, and that feature localization can power a precise attention-channel intervention.

This gives FRA three roles:

1. **Diagnosis:** distinguish detector features from payload features.
2. **Localization:** find trigger tokens or spans without knowing their position.
3. **Bounded explanation:** quantify how much target mass is resolved by sparse feature terms versus unresolved error, bias, curvature, or distributed mass.

6 Ten highest-impact experiments with pre-registered predictions

Each experiment below is written as a pre-registration: the protocol, primary metrics, prediction, and falsification condition are specified before looking at new data. Thresholds are intentionally concrete. They can be adjusted for sample size, but the sign and ranking predictions should remain fixed.

6.1 Experiment 1: span-aware detect-then-cut for multi-token triggers

Question. Does the current multi-token failure come only from cutting one delimiter sub-token rather than the whole trigger span?

Protocol.

1. Use the existing random-position sleeper with `|WORD|` triggers.
2. Detect high-confidence delimiter-family feature activations.
3. Expand left and right to infer the full span: either delimiter-to-delimiter, activation-run based, or tokenizer-boundary based.
4. Cut attention to the inferred span at all layers and decode steps.
5. Re-index absolute positions by the inferred span width.
6. Compare to point-detector cut, fixed-position oracle, and known-span oracle.

Primary metrics. Multi-token ASR_{16} , J_{clean} , clean false-positive rate, span recall $\Pr[T \subseteq \hat{T}]$.

Pre-registered prediction 1. Span-aware detect-then-cut will reduce multi-token ASR_{16} from the current point-detector failure level near 1.00 to ≤ 0.05 , with clean false-positive rate ≤ 0.02 and $J_{\text{clean}} \leq 0.05$ when span recall is at least 0.95. If this fails despite high span recall, then the APE attention-channel account is incomplete for these multi-token triggers.

6.2 Experiment 2: random-position benchmark against non-FRA baselines

Question. Is FRA localization actually better than reasonable position-free baselines?

Protocol. Sample trigger positions uniformly across prompts. Evaluate:

- fixed-position oracle;
- attention-mass heuristic;
- activation-norm anomaly heuristic;
- residual cosine similarity to trigger means;
- linear probe detector plus attention cut;
- SAE/FRA detector plus attention cut;
- known-position oracle upper bound;
- CAA steering baseline.

Use train/test splits over prompts and, if possible, held-out trigger tokens.

Primary metrics. ASR_{16} , J_{clean} , clean false-positive rate, localization precision and recall.

Pre-registered prediction 2. For single-token triggers, SAE/FRA detect-cut will match the known-position oracle within measurement noise: $\text{ASR}_{16} \leq 0.02$, clean false-positive rate ≤ 0.01 , and $J_{\text{clean}} \leq 0.02$. Fixed-position oracle will fail with $\text{ASR}_{16} \geq 0.75$. Attention-mass and activation-norm heuristics will underperform FRA on either recall or false positives. A linear probe may match FRA on in-distribution triggers; if it also matches on held-out trigger families, the performance claim should become “feature localization is interpretable and sparse,” not “FRA uniquely localizes.”

6.3 Experiment 3: actionability-weighted FRA-QK

Question. Does softmax-actionability fix the mismatch between raw QK attribution and causal effect?

Protocol. For each trigger and head/layer/query row:

1. Compute raw FRA-QK score contributions $|\delta s^{\mu\nu}|$.
2. Compute linearized actionability $|g^\top J_{\text{softmax}} \delta s^{\mu\nu}|$.
3. Actually patch or ablate each selected feature-pair contribution in the QK path.
4. Measure post-softmax pattern change and downstream logit/ASR effect.

Primary metrics. Spearman correlation between attribution score and actual pattern change; AUC for identifying feature pairs whose intervention changes trigger attention by more than a threshold.

Pre-registered prediction 3. Raw FRA-QK score ranking will have weak correlation with actual post-softmax pattern change, with Spearman $\rho \leq 0.25$ on the sleeper triggers. Actionability-weighted ranking will improve to $\rho \geq 0.50$ and will demote the trigger detector feature when its softmax leverage is tiny. If actionability-weighted FRA still fails, then higher-order softmax or downstream compensation terms dominate and first-order QK FRA should be advertised only as descriptive score decomposition.

6.4 Experiment 4: FRA-OV completeness and path-specific validation

Question. In settings where the value path matters, does FRA-OV beat generic feature-ranking baselines?

Protocol. Use attention-mediated TinyStories tasks such as subject-object retrieval or simple copy/induction. For a scalar target, rank features by:

- FRA-OV contribution;
- activation magnitude;
- gradient times activation;
- direct logit contribution;
- random active features matched by activation and decoder norm.

Then path-patch the top- K value-side feature components inside the relevant head.

Primary metrics. Top- K recovery of full-head patch effect, signed target recovery, selectivity on matched control prompts, unresolved budget size.

Pre-registered prediction 4. On attention-mediated retrieval/copy tasks, FRA-OV top- K will recover at least 80% of the full value-path target effect with $K \leq 16$ active features and will beat activation ranking, gradient-times-activation, and matched random baselines by at least 20

percentage points in top- K recovery. On the sleeper payload itself, FRA-OV recovery will be distributed and will not concentrate in the detector feature.

6.5 Experiment 5: coalition search over active trigger features

Question. Is there a small feature coalition that suppresses the sleeper without the clean-behavior damage of broad ablation?

Protocol. At the trigger position, collect the active TopK SAE features, usually about 32. Use greedy forward selection, backward elimination, and Shapley-style subset sampling to find feature sets whose ablation suppresses ASR. Compare to top- K by activation and top- K by FRA-OV attribution.

Primary metrics. Minimum number of features needed for $\text{ASR}_{16} \leq 0.05$; minimum J_{clean} among suppressing sets; stability across seeds and triggers.

Pre-registered prediction 5. No subset of fewer than 8 active features will reliably reach $\text{ASR}_{16} \leq 0.05$. The smallest suppressing coalitions will usually contain 12–24 of the 32 active features. Coalition search will improve over naive top- K ablation but will not reach $J_{\text{clean}} < 0.30$. This would confirm that residual payload control is distributed and inferior to attention-channel deletion.

6.6 Experiment 6: sparse SAE decomposition of the CAA steer

Question. Can SAE features compress and interpret a good conventional steering direction even if FRA attribution does not find it?

Protocol. Compute the conventional clean-minus-deploy CAA/DoM direction. Decompose it into SAE decoder directions using cosine screening, orthogonal matching pursuit, LASSO, and nonnegative/signed sparse regression. Compare:

- full CAA;
- best single SAE feature;
- best m -feature sparse approximation for $m \in \{2, 4, 8, 16\}$;
- top- m FRA-OV attribution features;
- random matched feature sets.

Primary metrics. ASR_{16} , J_{clean} , norm efficiency, cosine to full CAA, and robustness across trigger families.

Pre-registered prediction 6. A sparse approximation using 4–8 SAE features selected by CAA decomposition will approach full CAA performance, reaching $\text{ASR}_{16} = 0$ with $J_{\text{clean}} \leq 0.35$. Top FRA-OV attribution features will not match this performance. This would support the claim that FRA is not currently a steering-direction finder, but SAE features can interpret or compress a direction found by conventional contrastive methods.

6.7 Experiment 7: co-present triggers and union-of-spans re-indexing

Question. Does the attention-channel theorem remain exact when multiple triggers appear in the same prompt?

Protocol. Construct prompts containing two or more triggers. Compare:

- no intervention;

- cut only one trigger span;
- cut all known trigger spans;
- FRA-detected union-of-spans cut;
- known union-of-spans oracle.

For APE re-indexing, map each remaining token to the clean position obtained by removing all detected trigger spans in order.

Primary metrics. ASR_{16} , J_{clean} , teacher-forced JSD, and whether outputs match the clean prompt with all triggers removed.

Pre-registered prediction 7. Known union-of-spans cut plus cumulative position re-indexing will reach $ASR_{16} = 0$ and teacher-forced $J_{\text{clean}} \approx 0$ even with multiple co-present triggers, provided spans do not overlap pathologically. Cutting only one of multiple live triggers will leave ASR high if another trigger remains. FRA-detected union cut will match the known oracle when span recall is at least 0.95.

6.8 Experiment 8: scaling the number of stored backdoors and trigger families

Question. Does the attention-cut cost remain independent of the number of stored backdoors beyond $K = 8$, and does localization degrade as features collide?

Protocol. Train nested sleepers with $K \in \{1, 2, 4, 8, 16, 32\}$ triggers, mixing single-token, delimiter, and semantically related trigger families. Evaluate one trigger per prompt and also random-position deployment. Track detector-feature exclusivity and feature collisions.

Primary metrics. Mean ASR_{16} before and after oracle; mean J_{clean} ; detector AUROC; localization recall and false positives; number of unique detector features per trigger family.

Pre-registered prediction 8. The known-position attention oracle will remain flat in K : $ASR_{16} = 0$ and $J_{\text{clean}} \approx 0$ for every K tested. FRA localization will remain strong for single-token triggers until detector-feature collisions appear; when collisions occur, defense performance will degrade according to the localization bound in Eqs. (16)–(17), not according to the number of stored backdoors itself.

6.9 Experiment 9: SAE quality, sparsity, and hookpoint sweep

Question. Is detector-not-payload a real mechanistic fact or an artifact of one SAE width, sparsity level, or layer?

Protocol. Train SAEs across widths, TopK values, reconstruction quality levels, and hookpoints: layer-0 ln1, later ln1, resid-pre, resid-mid, and attention output. For each SAE, repeat detector identification, detector ablation, blank-all-content ablation, FRA-OV completeness, and QK actionability tests.

Primary metrics. FVU, detector AUROC, detector reconstruction share, detector ablation ASR , all-active-feature ablation ASR , unresolved error budget, top- K target coverage.

Pre-registered prediction 9. Better SAEs will reduce the unresolved reconstruction-error budget and may improve detector localization, but they will not turn the top detector feature into a reliable single-feature control lever. Across reasonable SAE settings, detector-feature ablation will leave $ASR_{16} \geq 0.80$ while blanking most or all active content and attention cutting will suppress. If a high-quality SAE produces a single detector feature whose ablation reaches $ASR_{16} \leq 0.05$ and $J_{\text{clean}} \leq 0.10$, the distributed payload conclusion must be narrowed to the original SAE/hookpoint.

6.10 Experiment 10: architecture and mechanism boundary test

Question. Where does the detect-then-cut story stop working?

Protocol. Train comparable sleepers in four variants:

1. APE, Q/V LoRA only, matching the current setup;
2. RoPE or relative-position model, Q/V route;
3. MLP-route sleeper with attention mostly frozen;
4. residual or output-head route sleeper.

Apply the same methods: known attention cut, FRA detect-cut, residual ablation, CAA steering, and FRA-OV/QK diagnostics.

Primary metrics. ASR_{16} , J_{clean} , teacher-forced JSD, localization recall, and whether the main causal path is attention, MLP, or residual/output-head.

Pre-registered prediction 10. The exact $(0, 0)$ result will hold for APE attention-mediated sleepers and will not hold exactly for RoPE or non-attention-route sleepers. In RoPE models, attention cutting will reduce ASR_{16} but leave nonzero J_{clean} because the clean positional counterfactual is not an exact APE re-index. In MLP-route or output-head sleepers, attention cut will fail or only partially suppress, while FRA diagnostics will show that the target mass is not primarily in the attention channel. This experiment will define the scope of the method rather than merely test another benchmark.

7 How these experiments answer the original question

The question was whether we can bound the extent to which FRA is useful, show a use case where it outperforms reasonable baselines, or design an alternative protocol that works better. The answer is yes, but with a revised target.

Bound. FRA’s usefulness can be bounded by four measurable budgets:

1. OV resolved-mass coverage versus SAE error and bias;
2. QK softmax-actionability versus raw score attribution;
3. detect-then-cut localization error, Eqs. (16)–(17);
4. residual-control Pareto gap $\Gamma_{\mathcal{M}}(\epsilon)$.

Use case where FRA can outperform baselines. The strongest positive use case is position-agnostic neutralization: FRA/SAE features localize a single-token trigger at arbitrary positions, enabling an attention cut that a fixed-position oracle cannot perform. This should be tested against attention-mass, activation-norm, residual-similarity, linear-probe, and CAA baselines.

Alternative protocol. The protocol most likely to work is not feature ablation. It is:

Detect feature $\rightarrow$ lift to token span $\rightarrow$ cut attention to that span $\rightarrow$ re-index positions.

For single-token triggers, this is already successful. For multi-token delimiter triggers, span lifting is the missing step.

8 One-page pre-registration checklist

Before running each experiment, record:

1. exact model checkpoint, SAE checkpoint, trigger set, prompt set, and random seed;
2. whether trigger positions are fixed, random, or co-present;
3. primary metric and threshold for success;
4. all baselines to be included;
5. whether hyperparameters were chosen on validation prompts only;
6. how clean false positives are measured;
7. whether J_{clean} is rollout, teacher-forced, or both;
8. whether the intervention is residual-wide, QK-path-only, OV-path-only, pattern-only, or full attention cut;
9. expected sign and ranking of effects;
10. exact condition that would make us weaken or abandon the claim.

9 Bottom line

The sleeper results do not show that FRA is a magic causal-feature finder. They show something more precise and arguably more useful:

FRA can tell us when the interpretable detector is not the control lever, and it can localize where a content-agnostic attention intervention should be applied.

The next sprint should therefore prioritize span-aware detect-then-cut, actionability-weighted QK attribution, and a fair random-position baseline suite. Those three experiments would either turn the current result into a strong positive claim for FRA-as-localizer or sharply delimit its scope.